

Situation

- In 1912 the large ship commonly known as the Titanic was sailing the North Atlantic Ocean when it collided into an iceberg.
- This incident required that passengers be rescued on lifeboats, which were in short supply.
- There were 2,240 passengers and crew on board.
- A subset of passenger data was recorded.
- We will focus on the child passengers.

Problem

- Not everyone could survive.
- How was it determined who gets an increased chance of survival?
- During the emergency some children survived, while others did not.
- Many factors (features) were recorded in for each passenger.

Question: Which factors affected the survival rate of children passengers during the sinking of the Titanic?

Factors in question: **PClass** (socio-economic class, 1=upper, 2=middle, 3=lower), **Sex** (male, female), **Age** (0-17), **SibSp** (number of siblings on board), **Parch** (number of parents/children on board), **Fare** (ticket cost), **Embarked** (region of where they embarked).

Solution

- **Pclass:** having a higher socioeconomic class (pclass = 1 or 2) could suggest an increased chance of survival.
- **Sex:** being female could suggest an increased chance of survival.
- **Age:** children aged 6 and under could suggest higher chance of survival.
- **SibSp:** having over 3 siblings could suggest decreased chance of survival.
- **Parch:** no insight on chance of survival.
- **Fare:** paying a fare over \$50 could suggest increased chance of survival.
- **Embarked:** embarking from Cherbourg (C) could suggest a higher chance of survival.

Next Steps

- Results suggested that child passengers who paid a fare over \$50 and had a higher socioeconomic class had an increased chance of survival.
- Boarding companies should implement laws to avoid biases favouring economic status.
Just because a passenger might be better off financially, does not mean their life should be put before someone else's.
- Moreover, boarding companies should not favour sex of passengers. In 1912, gender roles were more prominent in society, however both girls and boys should have equal chance of survival.
- These regulations should be implemented and monitored to determine whether their impact could increase overall survival, and decrease biases for future incidents.